POWER BI PROJECT REPORT

Laptop Sales Analysis Dashboard

Product Listing & Technical Insights

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Project Title	Laptop Sales Analysis Dashboard
Tool Used	Microsoft Power BI

1. Project Overview

The Laptop Sales Analysis Dashboard is a Power BI project created to explore a dataset of laptop listings and present useful information about laptop brands, prices, operating systems, processors, memory, storage, graphics and display specifications. The dashboard converts the tabular dataset into interactive visual summaries and product-level insights.

Dataset scope: The cleaned dataset used in the current Power BI report contains 1,020 rows and 29 columns. It represents laptop listing and specification information. The project focuses on analysing the available listing data; it should not be interpreted as verified units sold or revenue unless sales transaction data is added.

2. Objectives

- Create an interactive dashboard to summarise the laptop dataset.
- Compare laptop prices and key specifications across the available records.
- Identify the most frequently listed brands and commonly occurring product attributes.
- Explore technical specifications such as RAM, storage, processor, graphics and display.
- Present the findings in a clear, visually consistent format using multiple report pages.

3. Dataset Description

The project uses a cleaned CSV dataset imported into Power BI as the table laptop_cleaned2. The dataset contains 1,020 laptop records and 29 fields. The fields cover product identity, brand, price, rating, processor details, RAM, storage, graphics, display resolution, pixel density, touchscreen availability and operating system.

Field group	Example columns	Purpose
Product details	Name, Brand, Price, Rating	Identify listings and compare price/rating information.

Processor	Processor_brand, Processor_name, Processor_gen, Core_per_processor, Threads	Study processor types and computing specifications.
Memory & storage	RAM_GB, RAM_type, Storage_capacity_GB, Storage_type	Compare memory and storage configurations.
Graphics	Graphics_name, Graphics_brand, Graphics_GB, Graphics_integreted	Explore graphics hardware and memory.
Display & OS	Display_size_inches, Horizontal_pixel, Vertical_pixel, ppi, Touch_screen, Operating_system	Review display features and operating systems.

4. Data Cleaning Performed

A cleaned CSV file was prepared and imported into Power BI. The dashboard was built using the cleaned 29-column table. The available project context confirms the use of this cleaned dataset, but does not provide a complete step-by-step Power Query change log. The following points describe the preparation at a high level without assuming undocumented transformations:

- Used the cleaned dataset rather than the earlier raw, wider source file.
- Imported the cleaned CSV into Power BI and used the table name laptop_cleaned2.
- Used relevant product and specification columns for visuals and DAX measures.
- Excluded blank values from several 'most common' attribute calculations where applicable.

5. Data Model

The current report uses a single main table, laptop_cleaned2, containing the laptop listing records and their attributes. The dashboard measures and visuals refer to columns in this table. No separate dimension tables or table relationships were documented in the project details available for this report.

6. DAX Measures & Calculations

DAX measures were created to display summary values and the most frequently occurring attributes. The measures use the laptop table and calculate counts across its records.

Measure / insight	Column used	What it displays
Most Listed Brand	Brand	Brand appearing most often in the dataset.
Most Common OS	Operating_system	Most frequently listed operating system.
Most Used Processor	Processor_brand	Most frequent processor brand.
Most Common RAM	RAM_GB	Most frequently listed RAM capacity.
Most Common Graphics Brand	Graphics_brand	Most frequent graphics brand.
Most Common Screen Size	Display_size_inches	Most frequent display size.
Most Common Storage Type	Storage_type	Most frequent storage type.
Most Common Core Count	Core_per_processor	Most frequent CPU core count.
Most Common Thread Count	Threads	Most frequent thread count.
Most Common Graphics Memory	Graphics_GB	Most frequent graphics memory capacity.
Touchscreen Availability	Touch_screen	A text card label; confirm it is replaced by a data-driven measure if reporting actual prevalence.

Note: The final set of cards may differ depending on which measures were completed in Power BI. The touchscreen card discussed during development used the fixed text 'No Touchscreen'; it is not a calculated count or percentage of touchscreen models.

7. Dashboard Pages & Visualisations

Page 1 – Laptop Sales Overview

The first page provides an overview of the dataset using KPI cards and charts. The visuals developed for this page include summary cards, a donut chart, a line chart and a column chart for laptop counts by RAM capacity. The page uses a dark background with purple, cyan, pink and blue accent colours.

Page 2 – Technical / Product Analysis

The second page explores relationships and patterns across laptop specifications. The visuals discussed and developed include a ribbon chart, scatter plot, line chart and line-and-clustered-column chart. These visuals help compare product attributes and identify patterns in the dataset.

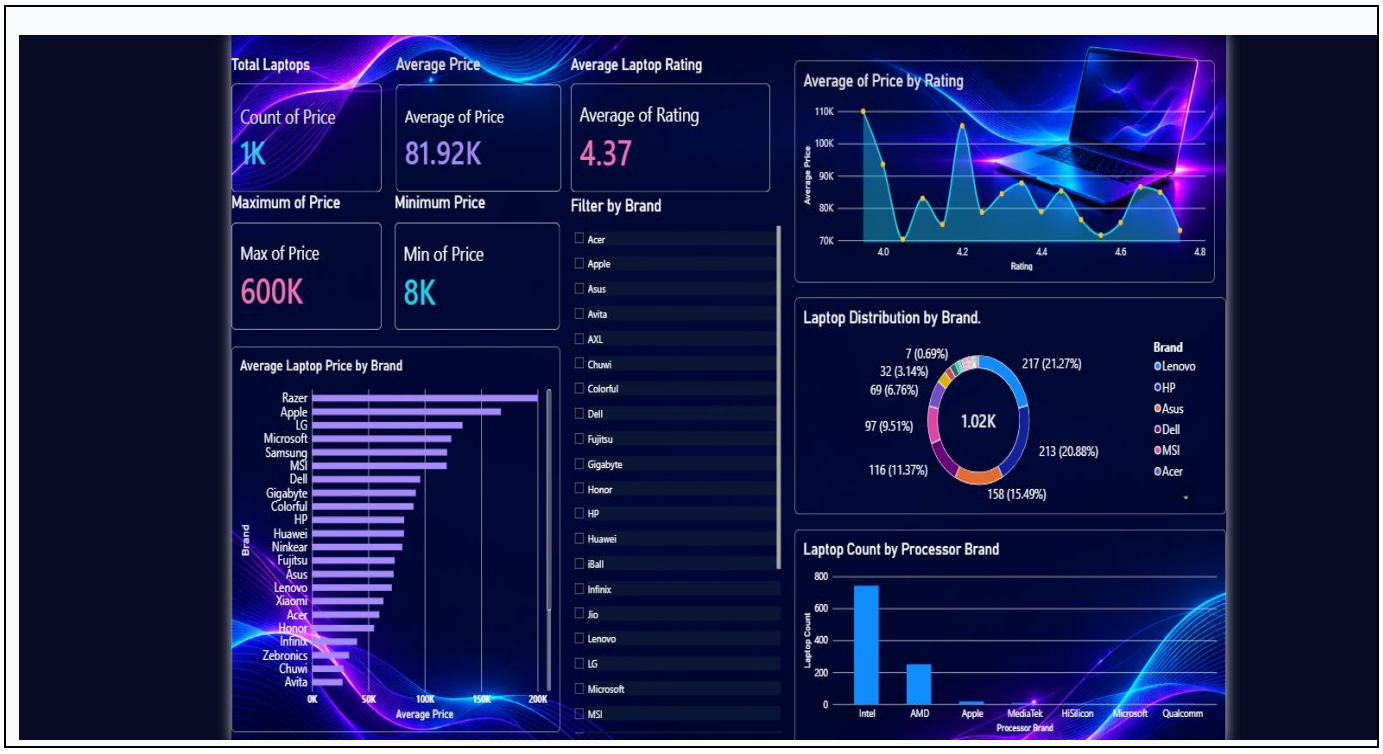
Page 3 – Product Insights

The third page presents product questions and key findings through card visuals rather than additional charts. The cards highlight frequently occurring attributes such as brand, operating system, processor, RAM, graphics brand, screen size, storage type and processor specifications.

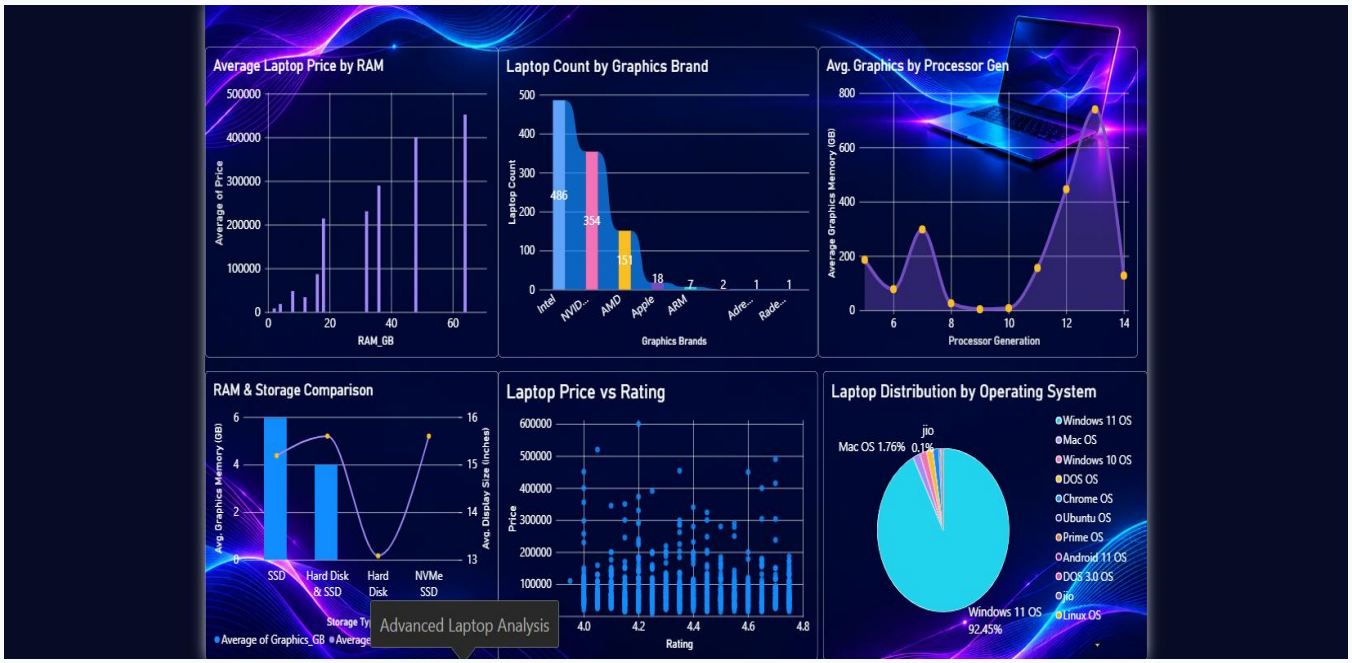
8. Dashboard Screenshots

Insert the final screenshots of the three Power BI report pages in the spaces below before publishing the report to GitHub. Screenshots were not included with the files provided for this report.

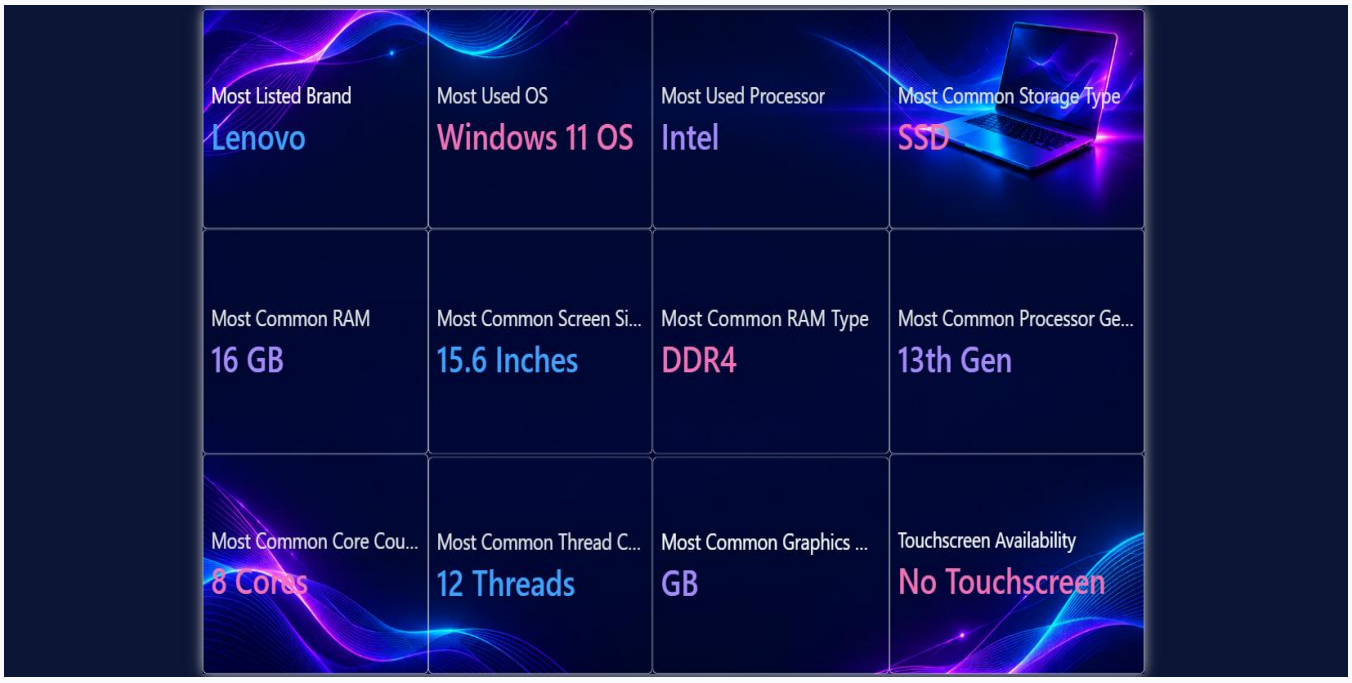
Page 1 – Laptop Sales Overview



Page 2 – Technical / Product Analysis



Page 3 – Product Insights



9. Key Findings

The dashboard is designed to surface the most common attributes and compare laptop specifications. Based on the dataset analysis available during development, examples of frequent values include:

- The most common CPU core count in the dataset is 8 cores.
- The most common thread count in the dataset is 12 threads.
- The most common graphics memory capacity in the dataset is 4 GB.
- The dataset contains 1,020 laptop records across 29 columns.

Other most-common values, such as brand, operating system, processor brand, RAM, graphics brand, screen size and storage type, should be copied from the final Power BI cards so the report exactly matches the finished dashboard.

10. Conclusion

The Laptop Sales Analysis Dashboard demonstrates how Microsoft Power BI can be used to transform a structured laptop dataset into an interactive and visually organised report. The three-page design combines an overall view, technical product analysis and card-based product insights. It helps users explore the distribution of laptop features and compare specifications in a clear way. The project can be extended by adding verified sales transaction data, more detailed data-quality documentation and final dashboard screenshots.